

SCRIPTING LANGUAGES (PYTHON)*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

Group-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): $2 \times 10 = 20$

i) Which of the following is not a valid way to import a module in Python?

- a) `import module_name`
- b) `from module_name import *`
- c) `include module_name`
- d) `import module_name as alias`

ii) What is the purpose of the "else" statement in an "if-else" construct in Python?

- a) It defines a second condition to check
- b) It defines a loop
- c) It specifies the code to execute when the condition in the "if" statement is false
- d) It raises an exception

iii) How do you define a multi-line comment in Python?

- a) Using the # symbol
- b) Using `'''` triple-quotes
- c) Using `//` symbol
- d) Using `/*` and `*/` symbols

iv) In Python, how do you remove an item from a list by its value?

- a) Using the `pop()` method
- b) Using the `remove()` method
- c) Using the `delete()` method
- d) Using the `discard()` method

v) What is the output of the code: `print(list(range(2, 6)))`?

- a) [2, 3, 4, 5]
- b) [2, 3, 4, 5, 6]
- c) [2, 4, 6]
- d) [2, 5]

vi) How do you define an empty dictionary in Python?

- a) `empty_dict = {}`
- b) `empty_dict = []`
- c) `empty_dict = ()`
- d) `empty_dict = set()`

vii) How do you check the length of a list in Python?

- a) `list.count()`
- b) `len(list)`
- c) `list.length()`
- d) `list.size()`

viii) How do you create a shallow copy of a list in Python?

- a) Using the `copy.copy()` method
- b) Using the `copy.deepcopy()` method
- c) Using the `list.copy()` method
- d) Shallow copies cannot be created in Python

ix) In Python, what does the `==` operator do?

- a) It assigns a value to a variable
- b) It checks if two values are equal
- c) It performs bitwise AND operation
- d) It concatenates two strings

- x) What is the main purpose of using functions in Python?
- To create loops
 - To organize code into reusable blocks
 - To define variables
 - To perform arithmetic operations
- xi) Which of the following loops is used to iterate over a sequence (such as a list or string) in Python?
- for loop
 - while loop
 - until loop
 - do-while loop
- xii) What is the correct way to create a list containing numbers from 1 to 5 in Python?
- list = [1, 2, 3, 4, 5]
 - list = (1, 2, 3, 4, 5)
 - list = {1, 2, 3, 4, 5}
 - list = range(1, 6)
- xiii) What is the purpose of the input() function in Python?
- It displays text on the screen
 - It imports external modules
 - It takes user input from the keyboard
 - It generates random numbers
- xiv) What does the import statement do in Python?
- It exports a variable to another module
 - It imports a module or package into the current script
 - It exports a module or package to another script
 - It defines a new function
- xv) What is the purpose of the join() method in Python when applied to strings?
- It splits a string into a list of substrings
 - It checks if two strings are equal
 - It concatenates multiple strings with a specified separator
 - It converts a string to lowercase

GROUP-B

- What is the purpose of indentation in Python?
 - What is the difference between a list and a tuple in Python?
 - What is an if statement, and how do you use it to make decisions in your code? 2+3+3
- Explain List slicing with examples.
 - What is Python Dictionary? Explain with an example. 4+4
- How a for loop is different from a while loop?
 - Write a program to get the sum of the digits of a number. 3+5
- Write short notes on the following arguments related to functions with examples:

 - Positional arguments
 - Default arguments
 - Variable-length arguments. 3+2+3
- Write different modes for opening a file with examples.
 - What is pattern matching? 6+2
- What is MVC?
 - What is Django and why should you use Django? 3+(2+3)
- Write a program to print elements of a tuple using a for loop.
 - What is the utility of a pass statement?
 - What is the utility of the else clause of a for loop? 4+2+2
- Differentiate between local and global variables with examples.
 - Show how to pass a list to a function using suitable code. 4+4

SCRIPTING LANGUAGES (PYTHON)

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A is compulsory and to be answered first. This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided. On early submission of answer scripts of Question No. 1, a student will get the remaining script earlier.

Answer any Five (05) Questions from Group B.

Group A

1. Choose the correct answer from the given alternatives (any twenty): 1x20

i) Which of the following is a Python tuple?
a) [1, 2, 3] b) (1, 2, 3) c) {1, 2, 3} d) {}

ii) What is the type of each element in sys.argv?
a) set b) list c) tuple d) string

iii) What will be the output of the follow Python statement? print("diploma"[2:-1])
a) plom b) pl c) iploma d) iplom

iv) Which one of the following is known as membership operator?
a) in b) not in c) both a and b d) is

v) Name the Python Library modules which need to be imported to invoke the following functions:
(i) sin() (ii) randint()
a) math b) random c) math & random d) None of them

vi) Identify the package manager for Python packages or modules.
a) Matplotlib b) python package c) plt.show() d) PIP

vii) Which is the correct form of declaration of dictionary?
a) Day = {1:'monday',2:'tuesday',3:'wednesday'}
b) Day = (1:'monday',2:'tuesday',3:'wednesday')
c) Day = [1:'monday',2:'tuesday',3:'wednesday']
d) None of them

viii) Identify the valid declaration of L: L = [1, 23, 'hi', 6].
a) tuple b) dictionary c) array d) list

ix) What will be the output of above Python code?
dict = {"abc":5, "def":6, "ghi":7}
print(dict[0])
a) abc b) 5 c) {"abc":5} d) Error

x) Which of the following is False?
a) capitalize() function in string is used to return a string by converting the whole given string into uppercase.
b) lower() function in string is used to return a string by converting the whole given string into lowercase.
c) String is immutable.
d) None of these.

xi) Which operator of the following has the lowest priority?
a) or b) & c) * d) **

xii) Slicing in python
a) returns a range of characters c) Both are correct
b) required start and end index d) None of them.

- xiii) Which of the following is valid arithmetic operator in Python?
a) : b)? c) < d) //
- xiv) Which of the following will give output as [13,24,91,751]? Where list1 = [6, 13, 3, 24, 0, 91, 8, 751]
a) print(list1[1:7:2]) c) print(list1[0:8:2])
b) print(list1[0:7:2]) d) print(list1[1:8:2])
- xv) Which of the following is True regarding lists in Python?
a) Lists are immutable.
b) Size of the lists must be specified before its initialization
c) elements can be of different data types.
d) None of them.
- xvi) What is the output of the following function call :
def func1(stdname, stdage = 30):
print(stdname, stdage)
func1('Neha', +10)
a) Neha 40 c) Neha 10
b) Neha 30 d) Error
- xvii) Modes of Python interpreters are
a) Interactive b) Script c) Both d) None
- xviii) What is the output of the following code :
import datetime
datetime = datetime.date(2022, 12, 18)
print(datetime)
a) Error c) 2022-12-18
b) 2022-18-12 d) 18-12-2022
- xix) What will be the output of the following Python code?
print(all([10, 8, 6, 4, 0]))
a) True b) False c) None d) Error
- xx) To return the length of string str1 what command do we execute?
a) str1.__len__() c) Both are correct
b) len(str1) d) None of them
- xxi) The elements of a list are arranged in descending order. Which of the following two will give same outputs?
A. list_name.sort()
print(list_name)
B. print(max(list_name))
C. list_name.reverse()print(list_name)
e)
D. print(list_name[-1])
a) A,B c) B,C
b) A,C d) C,D
- xxii) Write down the output of the following code.
s='a,b,c,d,'
print(s[1::2])
a) abcd c) bcd
b) bd d) ,,,
- xxiii) Which of the following items are present in the function header?
a) function name c) return value
b) parameter list d) Both a and b
- xxiv) All keyword in python are in
a) Lowercase c) Capitalized
b) Uppercase d) None of the above

xxv) What type of data is: arr = [(1,1),(2,2),(3,3)]?

- a) Array of tuples
- b) Tuples of lists

- c) List of tuples
- d) Invalid type

Group B

2. a) Define python. Explain the features of python programming.
b) Discuss the difference between local and global variable. (2+3)+3
 3. a) Explain the use of join() and split() string methods with examples.
b) What is Mutable and Immutable data type in python?
c) Describe why strings are immutable with an example. 3+2+3
 4. a) Explain break and continue statement with the help of for loop with an example.
b) Write a python program to check whether a number is prime or not. 4+4
 5. a) What is dictionary? Explain the methods available in dictionary.
b) Differentiate between the tuple and list in python. (2+2)+4
 6. a) State the purpose of using return statement with example in python function.
b) What do you mean by required and default argument which can be passed at the time of function call? 4+(2+2)
 7. a) Explain different types of operators with example of each.
b) Write down the example of importing a module in python. 4+4
 8. a) What do you mean module and package?
b) Explain the arbitrary arguments associated with function with example.
c) Write down the code to print current date and time. 2+3+3
 9. a) Describe the different access modes of the files with an example.
b) Discuss the following methods associated with the file object - i) read() ii) write(). 4+4
 10. a) What do you mean MVC framework? Explain each term.
b) Write down the steps of creating Django project and run it. 3+5
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SCRIPTING LANGUAGES (PYTHON)

307/2(N)

Time Allowed: 3 Hours

Full Marks: 60

Answer the following questions from Group-A, B & C as directed.**GROUP –A**

1. Choose the correct alternative (Any 10)

1 x 10=10

- i) Which one of the following is not a keyword in Python language? A) pass b) eval c) assert d) nonlocal
- ii) The input() function takes user's input as a - a) integer b) float c) string d) character
- iii) What is the output of the following?

```
x = ['ab', 'cd']
for i in x:
    x.append(i.upper())
print(x)
```

 - a) ['AB', 'CD']. b) ['ab', 'cd', 'AB', 'CD']. c) ['ab', 'cd']. d) none of the mentioned
- iv) Which statement is used to terminate the execution of the nearest enclosing loop in which it appears?
 - a) Pass b) break c) continue d) jump
- v) What will be the output of Python code snippet `x<<2` if `x=1`? a) 4 b) 2 c) 1 d) 8
- vi) The default access mode is a) r b) w c) rb d) wb
- vii) DRY principle makes the code a) Reusable b) Loop forever c) Bad and repetitive d) Complex
- viii) What will be the output of the following Python function?

```
min(max(False,-3,-4), 2,7)
```

 - a) -3 b) -4 c) error d) false
- ix) What will be the output of the Python expression `round(4.576)`? a) 4 b) 4.6 c) 5 d) 4.5
- x) Which function is called an anonymous function? a) Lambda b) Map c) Filter d) Reduce
- xi) What does pip stand for python? a) Unlimited length b) All private members must have leading and trailing underscores c) Preferred Installer Program d) None of these
- xii) Which of the following functions can help us to find the version of python that we are currently working on? a) `sys.version(1)` b) `sys.version(0)` c) `sys.version()` d) `sys.version`
- xiii) What will be the output of `print(10,20,30,sep="/")` a) 10 20 30 b) 10/20/30 c) 10\20\30 d) error
- xiv) Which of the following is a Python tuple? a) {1, 2, 3} b) {} c) [1, 2, 3] d) (1, 2, 3)
- xv) Predict output for the snippet

```
for j in range(1,5):
```


print(j)

a) 1 2 3 4 , b) indentation error, c) j j j j, d) j

j
j
j

2. Fill in the blanks (Any ten):

1 x 10=10

i) All keywords in Python are in _____.

ii) The Python built-in function for viewing variable types is _____.

iii) Django supports the _____ pattern.

iv) a = 5, b = 6

a, b = b, a

print(a,b)

_____ is the final output.

v) To convert '5' into int data type _____ command is used.

vi) Open() function is used for _____.

vii) a = 10 << 12

b = 34 >> 10, c = 7 | 13, d = 12 ^ 13

print(a,b,c,d)

_____ is the output.

viii) To find x^y , you will use _____ operator.

ix) A program in python is stored with a _____ extension.

x) The value of the expression `set([1, 1, 2, 3])` is _____.

xi) _____ method is used to delete a file.

xii) The _____ statement is a combination of an else statement and an if statement.

xiii) import time

a = time.localtime()

c = time.asctime(a)

print(c)

_____ is the output.

xiv) _____ is used to define a block of code in Python language.

xv) The `13//(4+3%5)` evaluates to _____.

3. Answer the following questions (any ten)

1 x 10 = 10

i) What is a lambda function?

ii) What is an Interpreted language?

iii) What is the purpose of `isatty()` ?

iv) What are python iterators?

v) What is Scope in Python?

vi) How to comment multiple lines in python?

vii) Explain `find()` function.

viii) How will you capitalize the first letter of string?

ix) What is the purpose of `flush()` ?

x) What are Python's dictionaries?

xi) How can you generate random numbers?

xiii) What is PEP 8?

xiv) What is the purpose of the `PYTHONPATH` environment variable?

xv) Is python a case-sensitive language?

GROUP –B

4. Answer the questions (Any six)

2x6=12

- i) What is slicing?
- ii) What are membership operators?
- iii) Differentiate between continue and break statement.
- iv) Differentiate between globals() and locals()
- v) Name two file built-in functions. Give their syntax.
- vi) What is the difference between Python Arrays and lists?
- vii) Give two ways to convert a number into a string.
- viii) B=3, A=2. Swap the values of these variables.
- ix) How can the ternary operators be used in python?
- x) How do you insert a static file to a template in Django?
- xi) Is Django better than Flask?
- xii) Write a Python code to add the values of two variables.

GROUP –C

5. Answer the question (any one)

6x1

- a) Explain the basic data types available in Python with examples.
- b) What is the difference between list and tuples in Python? What are the key features of Python?
- c) Write a Python program to print the factorial of a number.

6. Answer the question (any one)

6x1

- a) What is difference between module and package? Is pandas a module or package?
- b) Explain different types of loops available in Python with suitable examples.
- c) Write a python program to check whether the number given is a palindrome or not.

7. Answer the question (any one)

6x1

- a) What are input and output files? How do you use input and output files in Python?
 - b) Write a Python program to calculate the length of a string.
 - c) Write a python function to print multiplication table from 1 to 10.
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